

Project Walkthrough - Delhivery Graph-Based Network Intelligence

This document details the completed modules, technical decisions, and benchmark evaluations of the graph-based logistics intelligence system built for Delhivery.

Deliverables & Code Artifacts

- **Data Preprocessing & Graph Construction:** [build_graph.py](#)
- Processes `delivery_data.csv`, aggregates segment-level scans to OD-pair level (using training trips only to prevent leakage), and constructs a directed weighted NetworkX graph `logistics_graph.pkl`.
- **Centrality Audit & Visualization:** [audit_network.py](#)
- Computes degrees, unweighted betweenness centrality, and clustering coefficients. Ranks hubs by composite chokepoint/delay metrics and identifies delayed corridors. Saves tabular results and network chokepoints visualization.
- **Graph-Enhanced ETA Model:** [train_eta_model.py](#)
- Trains Node2Vec embeddings on the network graph and joins them with centrality features. Optimizes and evaluates a baseline OSRM-feature regressor versus the graph-enhanced LightGBM model.
- **FTL vs Carting Framework:** [route_type_framework.py](#)
- Quantifies FTL vs. Carting trade-offs and trains a Random Forest decision classifier for optimal route-type recommendation.
- **Interactive Streamlit Dashboard:** [app.py](#)
- A user-friendly, tabbed dashboard displaying benchmarks, bottleneck hub audits, delayed corridors, and an interactive route-type recommender.

- **Strategy Memo:** [strategy_memo.md](#)
- Executive strategy document for operational leadership summarizing top hubs, actions, and simulated revenue recovery.

Technical Validation & Benchmarking Results

1. Network Centrality Audit

Our audit revealed Gurgaon Bilaspur as the primary network chokepoint: - **Gurgaon Bilaspur HB (IND000000ACB)**: Betweenness Centrality = 0.2202 , Cumulative delay = 498,705 minutes, and SLA breach rate = 95.8% . - **Bangalore Nelamangala H (IND562132AAA)**: Betweenness Centrality = 0.1250 , Cumulative delay = 258,387 minutes, and SLA breach rate = 94.9% .

2. "Graph Advantage" in ETA Predictions

We benchmarked the models on the unseen test set (7,421 segments) with zero trip overlaps:

Metric	Baseline OSRM Model	Graph-Enhanced Model	Improvement
Mean Absolute Error (MAE)	46.23 minutes	35.81 minutes	22.54% reduction in error (10.42 mins saved)
SLA Prediction Accuracy	68.26%	75.69%	7.43% absolute increase (exceeding the 70% threshold)

The feature importance analysis proved that graph delay properties (src_total_delay_minutes , dest_total_delay_minutes) and Node2Vec embeddings (src_emb_14 , dest_emb_11) were selected as the most predictive features after OSRM estimations.

Launching the Streamlit Dashboard

You can run the interactive dashboard locally by executing:

```
streamlit run "c:\Users\Lenovo\Downloads\Graph-Based Network Intelligence\app.py"
```



This launches a browser-based application containing all network metrics, ranking tables, and the interactive route type recommendation tool.